

Audit of a proposed attack on (H6) and (H5)

withdrawn argument, retained with an explicit diagnosis

Abstract

This manuscript is retained as a record of an unsuccessful route. Its former main claim, “covariance-weighted Poincaré plus first moments implies (H6)”, is *not proved*. The induction uses the false inequality $\|v\|_\infty^2 \lesssim \|v\|_2^2$ on an infinite-dimensional L^2 space and treats the positive operator $M_u C M_u$ as trace class, whereas its trace diverges logarithmically because $C(0) = +\infty$. The precise counter calculations and a valid replacement under additional uniform one-point-density bounds are in `remaining.tex`, Propositions 6.1–6.2.

The proposed implication to the physical Hamiltonian gap is also not established. The Bauerschmidt–Dagallier inequality uses the unweighted lattice-gradient Dirichlet form of stochastic quantisation, not the covariance-weighted form assumed below, and the corresponding Markov generator is not the physical Hamiltonian. Uniform Euclidean-time decay for a dense family of observables would be an additional theorem.

The discussion correcting the proposed wavelet/Polchinski identification remains valid. All statements below that are labelled as a proposed claim or chain must be read as historical diagnosis, not as proved inputs to the spatial-cutoff theorem.

Withdrawal notice (8 August 2026)

Theorem 2.3 and the chain in §3 are withdrawn. In the proof of the former, lines corresponding to the estimate

$$\sum_{\alpha} c_{\alpha} \|v_{\alpha}\|_{\infty}^2 \lesssim \text{tr}(M_u C M_u)$$

have no valid justification: neither the displayed norm comparison nor the finiteness of the trace is true. The later transfer section is additionally blocked at its first premise because the cited LSI has a different Dirichlet form. See `remaining.tex` for the corrected proof program.

1 Setting, and the master hypothesis

Let C denote the time-zero free covariance $(2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$; its kernel is $C(x - y) = \frac{1}{2\pi} K_0(m_0|x - y|) > 0$, with $C \in L_{\text{loc}}^p$ for every $p < \infty$ and $C(0) = +\infty$ (logarithmic). Let φ_0 be the Gaussian measure with covariance C on $\mathcal{S}'(\mathbb{R})$, and $\mu_g = \Omega_g^2 d\varphi_0$ the ground-state measure. For real $u \in C_c^\infty$ put

$$F_n[u] := \int u(x) : \varphi(x)^n : dx,$$

Wick-ordered with respect to C , so that $V_B = \sum_n a_n F_n[h]$ for $P(\xi) = \sum a_n \xi^n$. Write ∇ for the Cameron–Martin gradient, $(\nabla F)(x) = \delta F / \delta \varphi(x)$, and $\mathcal{E}(F) := \mathbb{E}_{\mu_g} \langle \nabla F, C \nabla F \rangle$ for the Dirichlet form of the stochastic quantisation dynamics with invariant measure μ_g .

Hypothesis 1.1 (uniform Poincaré, (P-U)). There is $c_P < \infty$ with $\text{Var}_{\mu_g}(F) \leq c_P \mathbb{E}_{\mu_g} \langle \nabla F, C \nabla F \rangle$ for all $g \in \mathcal{G}$ and all F in the form domain.

(P-U) would be implied by a log-Sobolev inequality with this same covariance-weighted form. Bauerschmidt–Dagallier [1] do *not* provide that premise: their Dirichlet form is the unweighted lattice ℓ^2 gradient. Since the covariance multiplier tends to zero at high momentum, the unweighted inequality does not imply (P-U).

We also use, as an input, the uniform first-moment bound of Guerra–Rosen–Simon:

Lemma 1.2 ([3], Thm. 2). *For every $n \leq \deg P$ and every real u with $\text{supp } u \subseteq B$ there is $c_n(B)$, independent of g , with $|\mathbb{E}_{\mu_g}[F_n[u]]| \leq c_n(B) \|u\|_\infty$.*

Why the hypothesis of [3] Thm. 2 is met. Apply it with $W = \pm \varepsilon \int u : \xi^n$; one needs $P \pm \varepsilon \xi^n$ semibounded, which holds for $n < \deg P$ because the leading term of P dominates, and for $n = \deg P$ for ε small. Taking ω_g and using $H(g)\Omega_g = 0$ gives the bound. $\square$

2 Withdrawn claim: (P-U) $\Rightarrow$ (H6)

The two structural facts that make the induction work:

Lemma 2.1 (degree lowering). $\langle \nabla F_n[u], C \nabla F_n[u] \rangle = n^2 \iint u(x)u(y) C(x-y) : \varphi(x)^{n-1} : : \varphi(y)^{n-1} : dx dy$.

Proof. $(\nabla F_n[u])(x) = n u(x) : \varphi(x)^{n-1} :$, since $\frac{d}{d\varphi} : \varphi^n : = n : \varphi^{n-1} :$; insert into $\langle \cdot, C \cdot \rangle$. $\square$

Lemma 2.2 (positive definiteness). *For $k \geq 1$ the kernel $w_k(x, y) := u(x)u(y) C(x-y)^k$ is positive semidefinite. Consequently $w_k = \sum_\alpha c_\alpha v_\alpha \otimes v_\alpha$ with $c_\alpha \geq 0$, and for any $m \geq 0$*

$$\sum_\alpha c_\alpha \langle v_\alpha, C^{\circ m} v_\alpha \rangle = \text{tr}(W_k C^{\circ m}) = \iint u(x)u(y) C(x-y)^{k+m} dx dy < \infty,$$

where $C^{\circ m}$ denotes the kernel $C(x-y)^m$.

Proof. C is a covariance, hence positive definite; by the Schur product theorem the pointwise power $C^{\circ k}$ is positive definite; conjugating by multiplication by u preserves this. The trace identity is immediate, and the integral is finite because $C^{\circ(k+m)}$ is locally integrable in one dimension (only a logarithmic singularity, raised to a finite power) and u has compact support. $\square$

Theorem 2.3 (withdrawn claim; not established). *Assume (P-U) and Lemma 1.2. Then for every $n \geq 0$ there is $K_n < \infty$, depending only on n, c_P, P and $\text{supp } u$, such that*

$$\sup_{g \in \mathcal{G}} \mathbb{E}_{\mu_g} [F_n[u]^2] \leq K_n \langle u, C^{\circ n} u \rangle + c_n^2 \|u\|_\infty^2 \quad \text{for all real } u \in C_c^\infty(B). \quad (1)$$

In particular, taking $n = \deg P$ and $u = h$, (H6) holds: $\sup_g \|V_B \Omega_g\|^2 = \sup_g \mathbb{E}_{\mu_g} [V_B^2] < \infty$.

Invalid proof, retained to locate the failure. Induction on n .

$n = 0$: $F_0[u] = \int u$ is a constant; (1) is trivial.

$n = 1$: $\nabla F_1[u] = u$, so $\langle \nabla F_1, C \nabla F_1 \rangle = \langle u, C u \rangle$ is a constant and (P-U) gives $\text{Var}_{\mu_g}(F_1[u]) \leq c_P \langle u, C u \rangle$. With Lemma 1.2, $\mathbb{E}[F_1^2] = \text{Var} + (\mathbb{E} F_1)^2 \leq c_P \langle u, C^{\circ 1} u \rangle + c_1^2 \|u\|_\infty^2$.

$n - 1 \Rightarrow n$: By (P-U) and Lemma 2.1,

$$\text{Var}_{\mu_g}(F_n[u]) \leq c_P n^2 \mathbb{E}_{\mu_g} \left[\iint w_1(x, y) : \varphi(x)^{n-1} : : \varphi(y)^{n-1} : dx dy \right], \quad w_1 = u \otimes u \cdot C.$$

By Lemma 2.2 write $w_1 = \sum_{\alpha} c_{\alpha} v_{\alpha} \otimes v_{\alpha}$ with $c_{\alpha} \geq 0$. Since $\iint (v_{\alpha} \otimes v_{\alpha})(x, y) : \varphi(x)^{n-1} :: \varphi(y)^{n-1} : = F_{n-1}[v_{\alpha}]^2 \geq 0$, monotone convergence permits termwise expectation:

$$\text{Var}_{\mu_g}(F_n[u]) \leq c_P n^2 \sum_{\alpha} c_{\alpha} \mathbb{E}_{\mu_g}[F_{n-1}[v_{\alpha}]^2].$$

Apply the inductive hypothesis (1) at level $n - 1$ to each term and use Lemma 2.2 twice (with $m = n - 1$ for the first summand, $m = 0$ for the second, using $\|v\|_{\infty}^2 \lesssim \langle v, v \rangle$ on the fixed compact B):

$$\text{Var}_{\mu_g}(F_n[u]) \leq c_P n^2 \left[K_{n-1} \text{tr}(W_1 C^{\circ(n-1)}) + c_{n-1}^2 \kappa_B \text{tr}(W_1) \right] = c_P n^2 [K_{n-1} + c_{n-1}^2 \kappa'_B] \iint u u C^{\circ n},$$

which is $\leq \text{const} \cdot \langle u, C^{\circ n} u \rangle$, finite by Lemma 2.2. Adding $(\mathbb{E}_{\mu_g} F_n[u])^2 \leq c_n^2 \|u\|_{\infty}^2$ from Lemma 1.2 gives (1) at level n . $\square$

Remark 2.4 (what makes this work, and what it avoids). Three things. (i) ∇ lowers the Wick degree by exactly one, so the induction is finite and terminates at $n = 1$, where the Dirichlet form is a *constant*. (ii) The kernel produced by the Dirichlet form carries one extra factor of C , which is exactly what makes $\text{tr}(W_1 C^{\circ(n-1)}) = \iint u u C^{\circ n}$ finite — the logarithmic singularity of C is harmless at any finite power in one space dimension. (iii) Positive definiteness of $C^{\circ k}$ (Schur) turns the bilocal object into a positive combination of *squares*, so the inductive hypothesis applies termwise and no cancellation has to be controlled. Note that no correlation inequality, no sign of an Ursell function, and no reflection positivity is used — consistent with the diagnosis of [4, §6] that positivity-based tools stop at $:\varphi^2:$.

Remark 2.5 (the constant is not free-field-sharp). The bound is $K_n \langle u, C^{\circ n} u \rangle$, of the same form as the free-field value $\mathbb{E}_{\varphi_0}[F_n[u]^2] = n! \langle u, C^{\circ n} u \rangle$, but with K_n growing like $\prod_{j \leq n} c_P j^2$. Sharpness is not needed for M4.

3 Proposed, unproved chain to (H3) and (H5)

The following was the proposed chain. It is not a proof: step 1 is not derived for the measures and observables at issue, step 3 requires uniform decay for a dense family rather than only the two-point field, and the LSI generator is not the physical Hamiltonian.

1. *Uniform LSI $\Rightarrow$ exponential decay of correlations.* For lattice systems the equivalence of a volume-uniform log-Sobolev inequality with strong mixing (complete analyticity) is due to Stroock–Zegarliński; for unbounded continuous spins the relevant versions are Yoshida’s and Bodineau–Helffer’s. *Caveat:* this is the step whose hypotheses must be checked most carefully in the unbounded-spin setting, and it is not a formal consequence.
2. *No circularity.* [1] assume *bounded susceptibility*, i.e. $\int \langle \varphi(0), \varphi(x) \rangle dx < \infty$, which is strictly weaker than exponential decay. So “bounded susceptibility $\Rightarrow$ LSI $\Rightarrow$ exponential decay” is a genuine strengthening, not a loop.
3. *Euclidean decay $\Rightarrow$ mass gap.* By Feynman–Kac–Nelson, $\langle \varphi(0, x), \varphi(t, y) \rangle_{\nu_g} = \langle \Omega_g, \varphi(x) e^{-|t|H(g)} \varphi(y) \Omega_g \rangle$, so exponential decay in the Euclidean *time* direction at rate γ bounds the gap of $H(g)$ from below by γ on the subspace generated by the fields; since Ω_g is cyclic for the local algebra (local Fockness, [2]), this is the whole gap.

4. *Uniformity in N .* [1] is uniform in the lattice regularisation, so the decay rate, hence the gap, is uniform in the resolution: this is (H5).

4 A correction: wavelets are not a Polchinski decomposition

The strategy note proposed running the Bauerschmidt–Dagallier criterion “with the wavelet scale decomposition $C = \sum_j C_j$ in place of the Polchinski heat-flow decomposition”. This does not work as stated.

A Polchinski-type flow requires $C = \int_0^\infty \dot{c}_t dt$ with each increment $\dot{c}_t$ *positive semidefinite*; that is what makes the flow a genuine Gaussian convolution semigroup and what the Bakry–Émery-type argument integrates along. An orthogonal multiresolution analysis supplies projections Q_j and hence $C = \sum_j Q_j C Q_j + \sum_{j \neq k} Q_j C Q_k$: the cross terms vanish only if C is diagonal in the wavelet basis, which it is not (Daubechies wavelets almost-diagonalise $(-\Delta + m^2)^{-1}$, but “almost” is not enough for a positivity requirement).

The right object is the Brydges–Guadagni–Mitter *finite-range decomposition*: $C = \sum_j C_j$ with every C_j positive semidefinite *and* of finite range L^j . Such decompositions exist for $(-\Delta + m^2)^{-1}$ on $\mathbb{R}^d$ and on lattices, *uniformly in the lattice spacing* (Bauerschmidt). The wavelet resolution then enters not as the flow parameter but as the regularisation whose uniformity must be tracked — which is exactly the form in which [1] state their result. *This replaces Route β of strategy §7.2.*

5 Why the proposed transfer stops at its premise

Before any of the former transfer tasks below, one would need:

(T0) replace the unweighted Dirichlet form in [1] by the covariance-weighted form in (P-U), with constants uniform in the cutoff and regularisation.

No such implication follows by comparison of forms. Thus (T0) fails for the cited theorem, and the old tasks are retained only as a record of what would follow after a genuinely covariance-weighted estimate.

- (T1)** *Spatial cutoff instead of a box.* [1] treat finite volume with periodic or free boundary conditions; we need the coupling profile $0 \leq g \leq 1$. Since $g \leq 1$ pointwise, the interaction is dominated by the $g \equiv 1$ case. *To check:* that the hypotheses of the BD criterion (bounds on the effective potential along the flow) are monotone in the coupling profile.
- (T2)** *Infinite Euclidean time.* $\nu_g = \lim_{T \rightarrow \infty} \nu_{g,T}$ is an increasing limit of finite-volume measures; an LSI constant uniform in T passes to the limit. Routine.
- (T3)** *Marginalisation to time zero.* μ_g is the time-zero marginal of ν_g . An LSI is stable under pushforward by a linear projection, with the same constant, because $|\nabla_x F| \leq |\nabla F|$ for F depending on x only. *To check:* that the projected Dirichlet form is the one appearing in (P-U), i.e. that the 2-dimensional Cameron–Martin gradient restricted to sharp-time functionals reproduces $\langle \nabla F, C \nabla F \rangle$ with $C = (2\mu)^{-1}$.
- (T4)** *Regime.* [1] require bounded susceptibility, i.e. the high-temperature phase. By [5, Thm. 4.2] the uniform gap fails in a symmetry-broken phase anyway, so this restriction is not a loss: it coincides with the regime in which the programme can succeed at all.

6 Numerical validation, and milestones

(H6) is true in the solvable case

For $P(\xi) = \xi^2$ everything is Gaussian and exact: $\mathbb{E}_{\mu_g}[V_B^2] = [\int h (C_g - C)_{\text{diag}}]^2 + 2 \iint h h C_g^2$, with $C_g = (2\mu_g)^{-1}$. Numerically ($m_0 = 1$, $\lambda = 3$, so $m_1 = 2$; h a normalised Gaussian bump; $g = \mathbf{1}_{[-\ell, \ell]}$):

ℓ	first term	second term	$\mathbb{E}_{\mu_g}[V_B^2]$
0.5	0.00237	0.04812	0.05049
2	0.01109	0.03522	0.04631
4	0.01218	0.03499	0.04717
8	0.01219	0.03499	0.047175
16	0.01219	0.03499	0.047175
$g \equiv 1$	—	—	0.047175

The second moment saturates at the $g \equiv 1$ value and is stationary to all displayed digits beyond $\ell \approx 6$. Thus (H6) holds in this solvable quadratic example. A numerical example does not repair the withdrawn proof for general P .

Superseded milestones

- A** *Failed.* The spectral expansion exposes an infinite trace and cannot be repaired by domain bookkeeping. See `remaining.tex`, Proposition 6.1.
- B** (T3): the marginalisation lemma. This is the smallest self-contained piece and it is a prerequisite for everything else.
- C** *Blocked before (T1).* The form in [1] is unweighted, so (T0) fails.
- D** Replace the gap chain by the dense-family spectral criterion in `remaining.tex`, Lemma 5.1, and prove its Euclidean-decay premise separately.
- E** Numerical stress test: measure $\mathbb{E}_{\mu_g}[V_B^2]$ for *quartic* P by Monte Carlo on a lattice, as a function of ℓ and of the lattice spacing, and check saturation. A failure here would refute (H6) directly and cheaply.

References

- [1] R. Bauerschmidt and B. Dagallier, *Log-Sobolev inequality for the φ_2^4 and φ_3^4 measures*, Comm. Pure Appl. Math. **77** (2024) 2579–2612; arXiv:2202.02295.
- [2] J. Glimm and A. Jaffe, Acta Math. **125** (1970) 203–261.
- [3] F. Guerra, L. Rosen and B. Simon, Comm. Math. Phys. **27** (1972) 10–22.
- [4] *On (H6)*, `h6.tex`.
- [5] Milestone M3, `../m3/m3.tex`.